

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0510

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

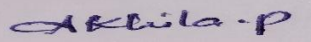
Assessment Tool Weightage: 10%

Code of Conduct:

I AKHILA POCHIMIREDDY(192425225) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>	<i>Marks(100)</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>	
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>	
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	
<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all</i>	<i>Mostly accurate with minor</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	

		<i>required points</i>	<i>omissions</i>			
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1. An image appears too bright.

Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

Answer

A very bright image contains many pixels with high intensity values (close to 255), causing loss of detail in bright regions. To reduce excessive brightness, **Power-Law (Gamma)**

Transformation with $\gamma > 1$ is the most suitable grey level transformation.

Definition

Grey level transformation modifies pixel intensity values to improve image quality without changing the image geometry.

Formula

$$s = cr^\gamma$$

Where:

- **r** = Input pixel intensity
- **s** = Output intensity
- **c** = Constant
- **γ** = Gamma value

For brightness reduction,

$$\gamma > 1$$

Working Process

- Read each pixel intensity.
- Apply gamma correction using γ greater than 1.
- High-intensity pixels are compressed.
- Bright areas become darker.
- Hidden details become visible.

Advantages

- Reduces overexposure.
- Improves visibility.
- Preserves image information.
- Easy to implement.

Applications

- Medical imaging
- Satellite images
- Industrial inspection

Expected Outcome

The image becomes balanced, overexposed regions are corrected, and object details become clearer.

2.A dark image lacks detail.

Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

Answer

Dark images contain low pixel intensity values, making important information difficult to observe. **Gamma Correction (Power-Law Transformation)** with $\gamma < 1$ enhances such images.

Formula

$$s = c r^\gamma$$

where $\gamma < 1$.

Working

- Read pixel values.
- Apply gamma correction.
- Low-intensity pixels are increased.
- Dark regions become brighter.
- Hidden details become visible.

Advantages

- Enhances dark areas.
- Maintains natural appearance.
- Preserves edges.
- Improves visibility.

Applications

- CCTV footage
- Medical images
- Night vision systems

Expected Result

Dark objects become clearly visible while preserving overall image quality.

3. An image has poor contrast.

Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

Answer

The best technique is **Histogram Equalization**.

Definition

Histogram Equalization is a contrast enhancement technique that redistributes pixel intensity values uniformly across the available intensity range.

Working Process

1. Compute histogram.
2. Calculate cumulative histogram (CDF).
3. Normalize CDF.
4. Map old intensity values to new intensity values.
5. Generate enhanced image.

Advantages

- Improves global contrast.
- Reveals hidden details.
- Uses full intensity range.
- Simple implementation.

Applications

- Medical diagnosis
- Satellite imaging
- Face recognition
- Machine vision

Result

The contrast increases and objects become easier to distinguish.

4.A low-contrast image requires uniform intensity distribution.

Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

Answer

The most suitable method is **Histogram Equalization**.

Concept

Histogram Equalization spreads pixel intensities uniformly over the entire grayscale range.

Procedure

- Calculate histogram.
- Find probability of each intensity.
- Compute cumulative distribution.
- Normalize cumulative values.
- Replace old pixel values.

Advantages

- Uniform intensity distribution.
- Better contrast.
- Improved image interpretation.
- Easy preprocessing step.

Applications

- MRI images
- X-rays

- Fingerprint recognition

Effect on Intensity Levels

Dark pixels become brighter and bright pixels are redistributed uniformly, improving visibility.

5. An image shows uneven brightness across regions.

Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

Answer

The suitable enhancement method is **Adaptive Histogram Equalization (CLAHE)**.

Definition

CLAHE improves local contrast by processing small image regions separately.

Working

1. Divide image into small blocks.
2. Perform histogram equalization on each block.
3. Limit excessive contrast.
4. Merge all enhanced blocks smoothly.

Advantages

- Corrects uneven illumination.
- Enhances local details.
- Prevents over-enhancement.
- Produces natural-looking images.

Applications

- Medical imaging
- Face detection

- Surveillance

Expected Result

Brightness becomes uniform and image quality improves significantly.

6.A noisy image contains random intensity variations.

Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

Answer

The most suitable smoothing filter is the **Mean Filter**.

Definition

The Mean Filter replaces each pixel with the average intensity of neighboring pixels.

Kernel

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Working

- Select a neighborhood.
- Calculate average intensity.
- Replace center pixel.
- Repeat for all pixels.

Advantages

- Removes Gaussian noise.
- Produces smooth images.
- Easy implementation.
- Fast processing.

Disadvantages

- Blurs edges.
- Reduces sharpness.

Applications

- Image preprocessing
- Medical images
- Camera systems

7. An image requires noise reduction without significant loss of details.

Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

Answer

The **Median Filter** is the best choice.

Definition

Median filtering is a nonlinear spatial filter that replaces the center pixel with the median value of neighboring pixels.

Working

- Select a 3×3 neighborhood.
- Arrange pixel values.
- Find the median.
- Replace center pixel.
- Repeat throughout image.

Advantages

- Removes salt-and-pepper noise.

- Preserves edges.
- Better than mean filtering for impulse noise.
- Maintains image quality.

Applications

- Medical imaging
- OCR
- Digital photography

Expected Result

Noise is removed while preserving important image details.

8. An image appears blurred after smoothing.

Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

Answer

The most suitable sharpening method is **Laplacian Filtering**.

Definition

The Laplacian operator enhances edges by detecting rapid intensity changes.

Formula

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Sharpened image:

$$g(x,y) = f(x,y) - \nabla^2 f(x,y)$$

Working

- Compute Laplacian.
- Detect edges.

- Add/subtract Laplacian from original image.
- Produce sharpened output.

Advantages

- Sharpens edges.
- Restores details.
- Improves visual quality.
- Easy implementation.

Applications

- Medical imaging
- Satellite images
- Industrial inspection

9.Fine details in an image are not clearly visible.

Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

Answer

A **High-Pass Filter** is used.

Definition

High-pass filtering enhances high-frequency components like edges and textures.

Kernel

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Working

- Apply convolution.
- Suppress low frequencies.
- Preserve high frequencies.
- Enhance fine image details.

Advantages

- Better edge visibility.
- Improves texture.
- Useful for feature extraction.
- Increases sharpness.

Applications

- Object detection
- OCR
- Face recognition
- Image enhancement

Result

Fine structures become more visible and easier to analyze.

10.A noisy image must be preprocessed before further analysis.

Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Answer

A suitable spatial filtering method is the **Median Filter**.

Definition

Median filtering removes impulse noise before image analysis.

Steps

1. Select neighborhood.
2. Sort intensity values.
3. Find median.
4. Replace center pixel.
5. Continue for all pixels.

Advantages

- Removes salt-and-pepper noise.
- Preserves edges.
- Improves segmentation accuracy.
- Enhances feature extraction.

Applications

- Medical image preprocessing
- Pattern recognition
- Object detection
- Machine vision

Conclusion

After preprocessing, the image becomes cleaner and more suitable for segmentation, edge detection, classification, and other computer vision tasks.